EX 4: Data Modelling and Form Design using Zoho Creator

Aim

To design a relational data model and construct a structured Vehicle Service Request Form application using Zoho Creator, incorporating standard basic and advanced fields, entity relationships, dynamic validation rules, formula calculations, auto-generated tracking identifiers, and reporting views.

Software & Hardware Required

- Software Platform: Zoho Creator (Low-Code Application Platform)
- Client Environment: Modern Web Browser with active Internet connection
- Application Scope: Vehicle Service & Repair Request Management

Data Modelling Architecture

Data modelling involves identifying core entities, grouping fields logically into structured form sections, and establishing relational mappings across forms.

Entity Relationship Mapping

Entity Name	Purpose / Function	Relational Binding
Vehicle Owner	Captures personal contact details and communication info.	One owner can submit multiple service requests.
Vehicle Model	Master entity storing model titles, standard	Master lookup referenced by service requests.

Entity Name	Purpose / Function	Relational Binding
	service packages, and durations.	
Service Request	Main transactional form storing service options, dates, and drop-off times.	Core form referencing Lookup and SubForm components.
Previous Service History	Tracks historic maintenance records, dates, and costs.	SubForm embedded within a request (1-to-Many).
Service Costing	Calculates estimated charges, discounts, and final payable amounts.	Dynamic formula calculation on request submit.
Service Tracking	Generates system request IDs, submission timestamps, and QR codes.	Auto-generated system fields attached to each record.

Form Field Specifications

The application uses a primary form (`Service_Request_Form`) configured with 13 Basic Fields and extended with 17 Advanced Fields.

Section A–D: Basic Fields Configuration (13 Fields)

Section	Field Name	Field Type	Configuration & Validation
A – Owner D...	Owner_Name	Name	Mandatory; First and Last name structure
A – Owner D...	Vehicle_Registration_Number	Single Line	Mandatory; Uppercase alphanumeric
A – Owner D...	Vehicle_Type	Dropdown	Mandatory; Options: Hatchback, Sedan, SUV, EV
B – Contact ...	Email_Address	Email	Standard email pattern validation
B – Contact ...	Contact_Number	Phone	Mandatory; Indian 10-digit (+91) format
B – Contact ...	Pickup_Address	Address	Complete address components
C – Vehicle &...	Odometer_Reading	Number	Numeric validation (> 0)
C – Vehicle &...	Service_Type	Radio	Mandatory; General, Major Repair, Express

Section	Field Name	Field Type	Configuration & Validation
C – Vehicle &...	Additional_Services_Requested	Multi Select	Alignment, Washing, Interior Detailing
C – Vehicle &...	Issue_Description	Multi Line	Detailed description text box
D – Scheduli...	Preferred_Service_Date	Date	Restrict past dates (Date >= Today)
D – Scheduli...	Preferred_Dropoff_Time	Time	Operating hours time picker
D – Authoriz...	Authorize_Estimate_And_Repair	Checkbox	Mandatory agreement check

Section E–H: Advanced Fields Configuration (17 Fields)

Section	Field Name	Field Type	Rules & Specifications
E – Vehicle V...	Vehicle_Model_Lookup	Lookup	Bound to Vehicle_Models master form
E – Vehicle V...	Vehicle_Damage_Photo	Image	Formats: JPG, PNG; Max file size: 2 MB

Section	Field Name	Field Type	Rules & Specifications
E – Vehicle V...	Insurance_Copy_Upload	File Upload	Formats: PDF, JPG, PNG; Max file size: 5 MB
E – Vehicle V...	Vehicle_Age_Years	Decimal	Numeric with decimal point support
F – Service H...	Previous_Service_History	SubForm	Grid table; Permissible rows: 0 to 10
F – Service H...	Insurance_Policy_Portal_URL	URL	Standard URL schema checking
F – Service H...	Vehicle_Issue_Video	Video	Max duration: 45 sec; Max size: 50 MB
F – Service H...	Engine_Sound_Voice_Note	Audio	Engine audio capture field
G – Cost & Pi...	Requires_Pickup_Drop_Service	Decision Box	Boolean selection (Yes/No)
G – Cost & Pi...	Estimated_Service_Cost	Currency	Currency in INR (₹); Value >= 0
G – Cost & Pi...	Service_Discount_Percent	Percent	Valid percentage range: 0% to 25%

Section	Field Name	Field Type	Rules & Specifications
G – Cost & Pi...	Net_Service_Cost_Payable	Formula	Dynamic read-only calculated field
H – Authoriz...	Service_Center_Guidelines_Display	Rich Text	Read-only policy reference text block
H – Authoriz...	Owner_Digital_Signature	Signature	Mandatory touch/mouse signature pad
H – System V...	Service_Request_Submitted_On	Date-Time	System submission timestamp
H – System V...	Service_Request_ID	Auto Number	Format: SVC-REQ-2026- + 6-digit sequence
H – System V...	Service_Tracking_QR_Code	QR/Barcode	Encodes generated Service_Request_ID

1. Step-by-Step Procedure

Step 1: Create Application & Core Form

1. Log into **Zoho Creator** and click **Create New Application**.

2. Select **Create from Scratch** and enter the name `Vehicle Service & Repair Request Management`.
3. Create the initial form titled `Service_Request_Form`.

Step 2: Form & Relational Data Modelling

1. Identify transactional and reference entities: Owner, Vehicle, Model Master, Service Log, Pricing, and System Tracking.
2. Structure the fields into logical UI sections: *Owner Details*, *Contact Details*, *Vehicle & Issue*, *Scheduling*, *Vehicle Verification*, *Service History*, *Cost & Pickup*, and *Authorization / System*.

Step 3: Setup Master Form & Lookup Field

1. Create a separate master form named `Vehicle_Models`.
2. Add fields: `Model_Name` (Single Line), `Standard_Service_Package` (Dropdown), and `Estimated_Duration_Hours` (Decimal).
3. Populate sample records (e.g., *Hyundai i20*, *Honda City*, *Tata Nexon*).
4. Reopen `Service_Request_Form`, drag a **Lookup** field (`Vehicle_Model_Lookup`), and link it to `Vehicle_Models`.

Step 4: Basic Fields Setup & Validation Rules

1. Add the 13 Basic Fields into Sections A through D as mapped in the specification table.
2. Toggle mandatory status (`Is Mandatory` = True) for `Owner_Name`, `Vehicle_Registration_Number`, `Vehicle_Type`, `Contact_Number`, `Service_Type`, and `Authorize_Estimate_And_Repair`.
3. Enable email verification on `Email_Address` and set country format to India for `Contact_Number`.
4. Add field constraints: Set `Odometer_Reading` minimum value to 1 and specify `Preferred_Service_Date` validation rule to prevent past date selection.

Step 5: Advanced Fields Integration & SubForm Configuration

1. Place the 17 Advanced Fields into Sections E through H.
2. Insert a **SubForm** field named `Previous_Service_History`.
3. Configure internal SubForm fields: `Service_Date` (Date), `Service_Type` (Dropdown), and `Service_Cost` (Currency).
4. Set SubForm row limits between `0` and `10` rows.

Step 6: Pricing Formula Setup

2. Place a **Formula** field named `Net_Service_Cost_Payable` in Section G.
3. Configure the return type as Currency (INR).
4. Insert the computation formula:

$$\text{Net Cost} = \text{Estimated Cost} - \left(\text{Estimated Cost} \times \frac{\text{Discount %}}{100} \right)$$

4. Mark `Net_Service_Cost_Payable` as **Read-Only** so it recalculates automatically whenever `Estimated_Service_Cost` or `Service_Discount_Percent` changes.

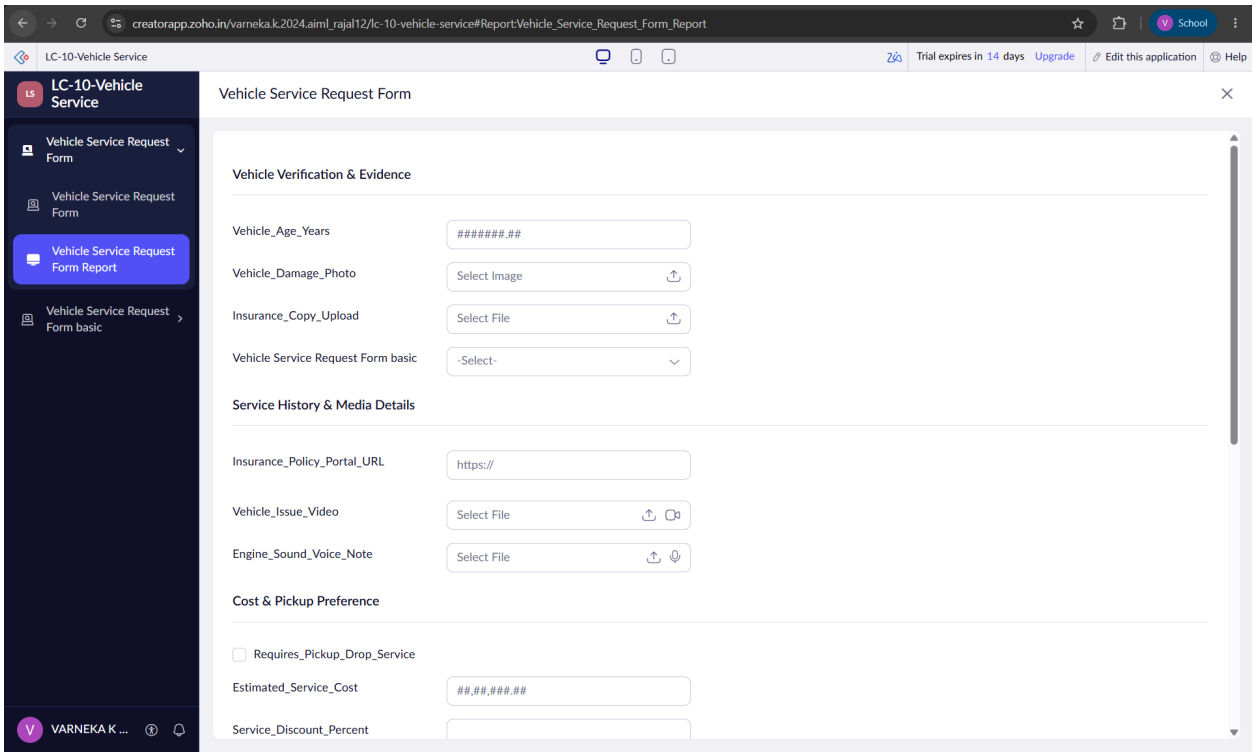
Step 7: System Identifiers, QR Code & Media Constraints

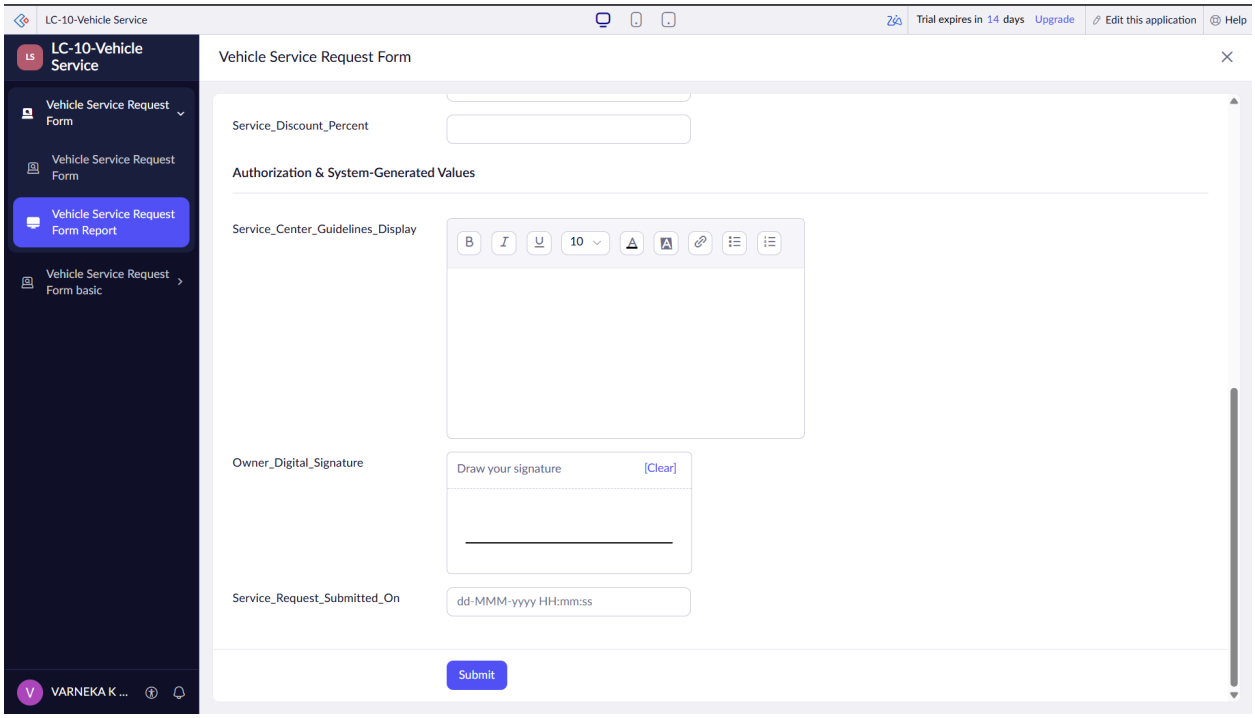
1. Set `Service_Request_ID` as Auto Number with prefix `SVC-REQ-2026-` and 6 digits.
2. Set `Service_Request_Submitted_On` to capture current Date-Time automatically upon submission.
3. Configure `Service_Tracking_QR_Code` to reference the value of `Service_Request_ID`.
4. Enforce media upload constraints:
    - **Insurance Copy:** PDF/JPG/PNG up to 5 MB.
    - **Damage Photo:** JPG/PNG up to 2 MB.
    - **Issue Video:** Duration <= 45 seconds, size <= 50 MB.
    - **Discount Percent:** Enforce range 0% to 25%.

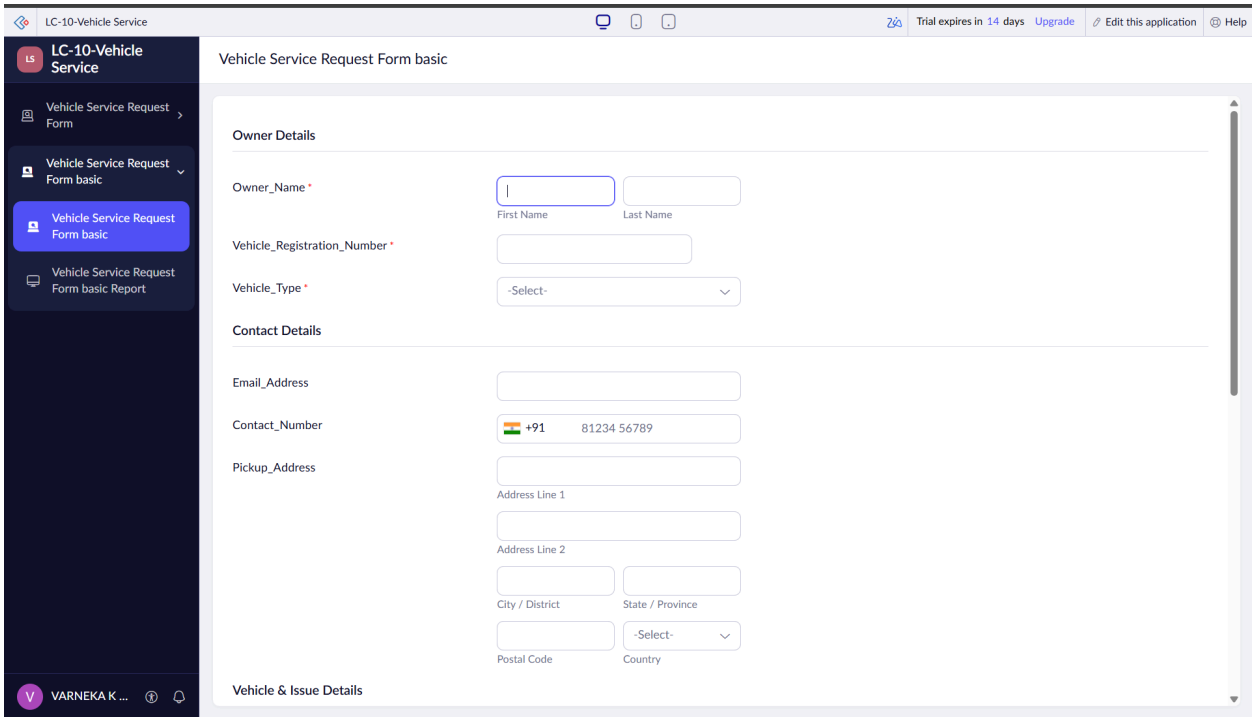
Step 8: Report Generation & Verification

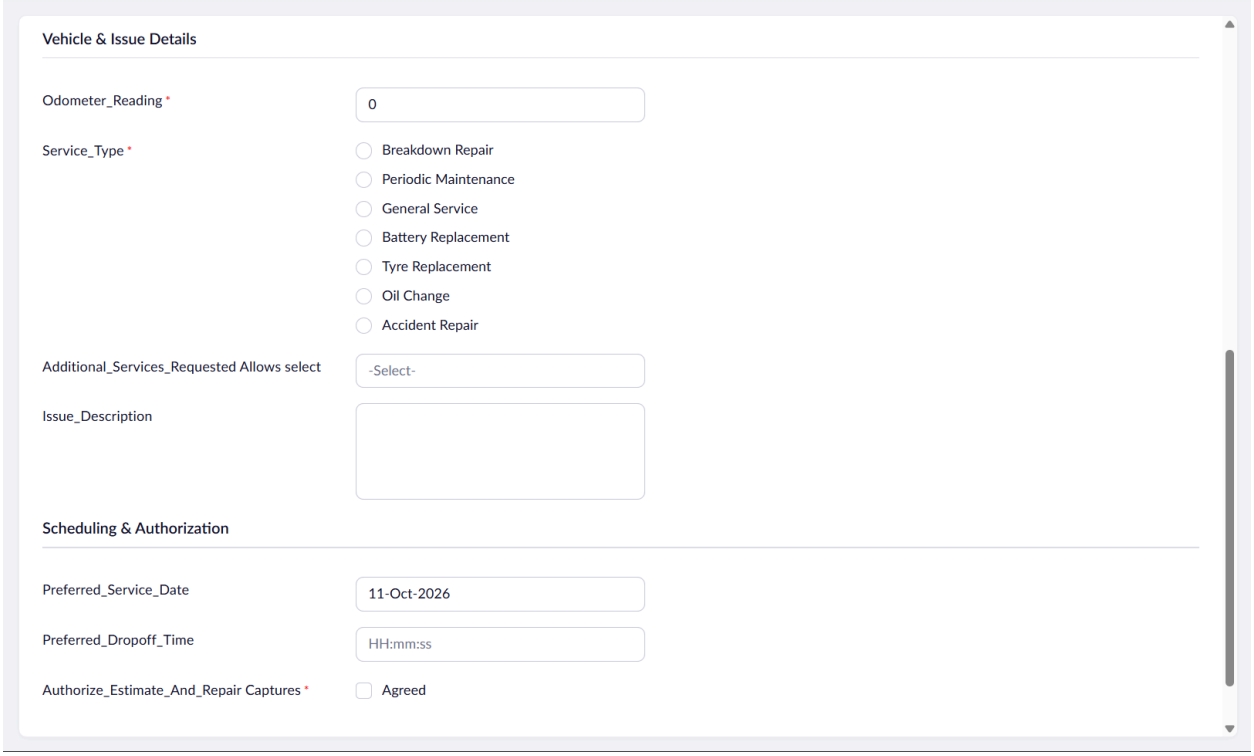
1. Access **Reports** in Zoho Creator and build `Service_Request_Report`.
2. Configure report display columns: `Service_Request_ID`, `Owner_Name`, `Vehicle_Registration_Number`, `Service_Type`, `Net_Service_Cost_Payable`, `Service_Request_Submitted_On`, and `Service_Tracking_QR_Code`.
3. Submit test records via the form to verify Lookup selection, SubForm entries, formula precision, media rules, and QR rendering.

Screenshots




LC-10-Vehicle Service
Trial expires in 14 days
Upgrade
Edit this application
Help
LC-10-Vehicle Service
Vehicle Service Request Form
Vehicle Service Request Form
Vehicle Service Request Form
Vehicle Service Request Form Report
Vehicle Service Request Form basic
Service_Discount_Percent
Authorization & System-Generated Values
Service_Center_Guidelines_Display
Owner_Digital_Signature
Draw your signature
[Clear]
Service_Request_Submitted_On
dd-MMM-yyyy HH:mm:ss
Submit
VARNEKA K ...


LC-10-Vehicle Service
Trial expires in 14 days
Upgrade
Edit this application
Help
LC-10-Vehicle Service
Vehicle Service Request Form basic
Vehicle Service Request Form
Vehicle Service Request Form basic
Vehicle Service Request Form basic
Vehicle Service Request Form basic Report
Owner Details
Owner_Name *
First Name
Last Name
Vehicle_Registration_Number *
Vehicle_Type *
-Select-
Contact Details
Email_Address
Contact_Number
+91
81234 56789
Pickup_Address
Address Line 1
Address Line 2
City / District
State / Province
Postal Code
-Select-
Country
Vehicle & Issue Details
VARNEKA K ...



Result

The **Vehicle Service Request Management** application was successfully built and verified in Zoho Creator using structured data modelling principles. Form validations ensure accurate input, Lookup and SubForm relationships maintain dynamic data linkage, and automated formula computations and auto-generated QR identifiers enable seamless service tracking.